

Auto Scene Assessment Agent

Tier 2 Computer Vision Agent Using YOLO & CLIP

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ITAI 1378 - Computer Vision
Final Project

Perceive -> Reason -> Act

What I built

For my final project, I built a computer vision agent that uses **two different CV tools** to understand an image before making a decision.

The goal:

Combine object detection and scene understanding instead of relying on one model alone.

My Agent Uses:

- YOLO → detects objects
- CLIP → identifies the overall scene
- Rule-based logic → compares the results and decides what to do
- Human review → handles uncertainty and model disagreement

Tier 2: One agent + multiple computer vision tools

How My Agent Works

System Architecture

IMAGE INPUT



YOLO

What objects are here?



CLIP

What kind of scene is this?



RULE-BASED AGENT

Do the results make sense together?



FINAL DECISION



SAVE RESULTS + TRACE

The agent follows a **Perceive** → **Reason** → **Act** workflow.

Tool #1: YOLO

YOLO: Object Detection

YOLO gives my agent **object-level information** about each image.

YOLO returns:

- Object labels
- Confidence scores
- Bounding boxes
- Object count
- Inference time

Example: Airport Image



YOLO detected:

- Multiple people
- Backpack
- **12 total detections**

YOLO answers: “What objects can I see?”

Tool #2: CLIP

CLIP: Scene Understanding

CLIP looks at the **whole image** instead of detecting individual objects.

It compares the image to possible scene descriptions and chooses the best match.

Examples from my testing:

Image	CLIP Result	Confidence
Sunset	Sunset landscape	99.9%
Airport	Airport scene	100%
Car lot	Parking lot with cars	97.3%
Morocco market	Busy outdoor market	99.3%

CLIP answers: “What kind of scene am I looking at?”

The Agent Makes the Decision

Rule-Based Agent Reasoning

This is what turns the two CV models into an **agent**.

The agent compares:

CLIP scene + CLIP confidence + YOLO objects

Then it decides whether the models support each other.

Examples:

Airport + people/backpack

→ **TRAVEL SCENE DETECTED**

Parking lot + multiple cars

→ **VEHICLE AREA DETECTED**

Dog scene + dogs

→ **ANIMAL SCENE DETECTED**

Low confidence

→ **MANUAL REVIEW RECOMMENDED**

The agent does not automatically trust every prediction.

Testing the System

Testing My Agent

I started with **5 test images** and then added **5 more challenging images**.

Final Test Set: 10 Images

The images included:

- Airport
- Sunset
- Parking lot
- Dogs
- Morocco market
- Traffic
- Downtown
- Cars at night
- Outdoor scene
- Rain

Why I added more images:

The second test set helped me see how the system handled **different lighting, environments, uncertainty, and harder scenes**.

Final Results

Evaluation Results

Make these numbers BIG on the slide:

10

Test Images

10/10

Successful End-to-End Runs

100%

Pipeline Completion

4

Cases Flagged for Review

40%

Review Rate

0.212 sec

Average YOLO Inference Time

At the bottom:

The pipeline successfully processed every test image, even when the models made imperfect predictions.

This should be one of your cleanest slides.

When the Models Got It Wrong

Failure Cases Were Actually Useful

This is one of the strongest parts of your project.

Cars at Night

CLIP:
Road/traffic scene — 81.6%

YOLO:
Wine glass, Wine glass 🤔

Agent decision:

POSSIBLE TRAFFIC SCENE – REVIEW

Then beside it:

Outdoor Image

CLIP:
Dog scene — 77.0%

YOLO:
Bench + potted plant

Agent decision:

MODEL DISAGREEMENTS - REVIEW



A high confidence score does not always mean the prediction is correct.

Human-in-the-Loop

Why Human Review Matters

One thing this project made really clear to me is that an AI system should be able to recognize when its answer might not be reliable.

My agent flags cases when:

- CLIP confidence is low
- YOLO does not support the scene
- The two models disagree

Instead of forcing a decision:

UNCERTAIN RESULT → HUMAN REVIEW

Human-in-the-loop is not just a backup. It is part of making an AI system more responsible and useful.

What I Learned From This Project

1. Multiple models can see the same image differently.

YOLO and CLIP gave me two completely different types of information.

2. Confidence $\neq$ correctness.

Some of my most interesting results were confident but wrong.

3. Testing failures helped improve the agent.

Instead of deleting bad results, I used them to create review rules.

4. Debugging is part of building the system.

I had to inspect data structures, fix errors, rerun cells, and verify outputs.

5. Human-in-the-loop matters.

Sometimes the best decision an AI system can make is admitting that a person should review the result.

Final System:

YOLO + CLIP + Rule-Based Reasoning + Human Review